Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com/. Reproduction and distribution is prohibited.

BS ISO 55000:2024

BSI Standards Publication

Asset management — Vocabulary, overview and principles

bsi.

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

National foreword

This British Standard is the UK implementation of ISO 55000:2024. It supersedes BS ISO 55000:2014, which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee AMS/1, Asset management.

A list of organizations represented on this committee can be obtained on request to its committee manager.

Contractual and legal considerations

This publication has been prepared in good faith, however no representation, warranty, assurance or undertaking (express or implied) is or will be made, and no responsibility or liability is or will be accepted by BSI in relation to the adequacy, accuracy, completeness or reasonableness of this publication. All and any such responsibility and liability is expressly disclaimed to the full extent permitted by the law.

This publication is provided as is, and is to be used at the recipient's own risk.

The recipient is advised to consider seeking professional guidance with respect to its use of this publication.

This publication is not intended to constitute a contract. Users are responsible for its correct application.

© The British Standards Institution 2024
Published by BSI Standards Limited 2024

ISBN 978 0 539 17975 0

ICS 01.040.03; 03.100.01

Compliance with a British Standard cannot confer immunity from legal obligations.

This British Standard was published under the authority of the Standards Policy and Strategy Committee on 31 July 2024.

Amendments/corrigenda issued since publication

Date	Text affected

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

BS ISO 55000:2024

International Standard

ISO 55000

Asset management — Vocabulary, overview and principles

Gestion d'actifs — Vocabulaire, aperçu général et principes

Second edition
2024-07

Reference number
ISO 55000:2024(en)

© ISO 2024

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

COPYRIGHT PROTECTED DOCUMENT

© ISO 2024

All rights reserved. Unless otherwise specified, or required in the context of its implementation, no part of this publication may be reproduced or utilized otherwise in any form or by any means, electronic or mechanical, including photocopying, or posting on the internet or an intranet, without prior written permission. Permission can be requested from either ISO at the address below or ISO's member body in the country of the requester.

ISO copyright office
CP 401 • Ch. de Blandonnet 8
CH-1214 Vernier, Geneva
Phone: +41 22 749 01 11
Email: copyright@iso.org
Website: www.iso.org

Published in Switzerland

Contents

Page

Foreword ..... **iv**
Introduction ..... **v**
1 Scope **1**
2 Normative references **1**
3 Terms and definitions **1**
3.1 Terms relating to assets 1
3.2 Terms relating to asset management 3
3.3 Terms relating to asset management system 4
4 Overview of asset management **8**
4.1 General 8
4.2 Principles of asset management 9
4.2.1 General 9
4.2.2 Value 9
4.2.3 Alignment 9
4.2.4 Leadership 9
4.3 Outcomes and benefits from asset management 9
4.4 Relationship between asset management, asset management system and assets 10
4.5 Asset management improvement and maturity 11
Bibliography **13**
Index **14**

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Foreword

ISO (the International Organization for Standardization) is a worldwide federation of national standards bodies (ISO member bodies). The work of preparing International Standards is normally carried out through ISO technical committees. Each member body interested in a subject for which a technical committee has been established has the right to be represented on that committee. International organizations, governmental and non-governmental, in liaison with ISO, also take part in the work. ISO collaborates closely with the International Electrotechnical Commission (IEC) on all matters of electrotechnical standardization.

The procedures used to develop this document and those intended for its further maintenance are described in the ISO/IEC Directives, Part 1. In particular, the different approval criteria needed for the different types of ISO document should be noted. This document was drafted in accordance with the editorial rules of the ISO/IEC Directives, Part 2 (see www.iso.org/directives).

ISO draws attention to the possibility that the implementation of this document may involve the use of (a) patent(s). ISO takes no position concerning the evidence, validity or applicability of any claimed patent rights in respect thereof. As of the date of publication of this document, ISO had not received notice of (a) patent(s) which may be required to implement this document. However, implementers are cautioned that this may not represent the latest information, which may be obtained from the patent database available at www.iso.org/patents. ISO shall not be held responsible for identifying any or all such patent rights.

Any trade name used in this document is information given for the convenience of users and does not constitute an endorsement.

For an explanation of the voluntary nature of standards, the meaning of ISO specific terms and expressions related to conformity assessment, as well as information about ISO's adherence to the World Trade Organization (WTO) principles in the Technical Barriers to Trade (TBT), see www.iso.org/iso/foreword.html.

This document was prepared by Technical Committee ISO/TC 251, *Asset management*.

This second edition cancels and replaces the first edition (ISO 55000:2014), which has been technically revised.

The main changes are as follows:

— the document has been restructured to provide an insight and understanding of asset management and its principles, its outcomes and benefits, the relationship between asset management, an asset management system and the assets of an organization, and indicators of asset management maturity;

— the principles of asset management have been revised;

— the outcomes of asset management have been introduced;

— the benefits have been rephrased and extended;

— the explanation of the elements of an asset management system has been deleted;

— the integration of management systems has been introduced;

— the maturity of an asset management organization has been introduced;

— the annexes have been deleted.

Any feedback or questions on this document should be directed to the user's national standards body. A complete listing of these bodies can be found at www.iso.org/members.html.

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Introduction

0.1 Purpose

This document provides an overview of asset management, its principles, and the outcomes and expected benefits from adopting asset management, and introduces the asset management system and relevant terminology.

This document sets the context for ISO 55001 and ISO 55002, and also later documents in the ISO 55000 family of standards.

0.2 Relationship with other standards in the ISO 55000 family of standards

ISO 55001 specifies requirements for an asset management system. ISO 55002 gives guidance on the design and application of the asset management system.

This document, ISO 55001 and ISO 55002 can be used in combination with relevant sector-specific or asset-type-specific standards and technical specifications to achieve the objectives of asset management.

Other standards on asset management developed by ISO/TC 251 can be used in the context established by this document, ISO 55001 and ISO 55002, including:

— ISO/TS 55010, which provides guidance on aligning financial and non-financial functions in asset management, and promoting understanding, implementation and improvement of this alignment for organizational benefits;

— ISO 55011, which offers guidance for developing public policy to promote asset management, focusing on the external context of organizations responsible for assets, with the aim of creating an enabling environment through consistent public policy instruments;

— ISO 55012, which focuses on people involvement and competence within an asset management system, offering guidance to enhance personnel commitment, effectiveness, knowledge and awareness, and promoting continual improvement;

— ISO 55013, which provides guidance on managing data assets, helping organizations increase and sustain the usefulness of data assets to meet asset management and organizational objectives.

0.3 Target audience

This document is primarily intended for use by:

— senior executives who wish to adopt asset management and implement an asset management system in their organization;

— those involved in asset management or intending to become practitioners;

— anyone who would benefit from a short introduction to asset management.

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Asset management — Vocabulary, overview and principles

1 Scope

This document defines terms and establishes principles and outcomes for asset management. It describes:

— the benefits of asset management and an asset management system;

— the relationship between asset management, the asset management system and asset portfolio;

— asset management improvement and maturity.

This document is applicable to all types of assets and all types and sizes of organizations.

This document does not provide financial, accounting, human resources nor technical guidance for managing specific asset types.

NOTE For the purposes of this document, ISO 55001 and ISO 55002, the term "asset management system" is used to refer to a management system for asset management.

2 Normative references

There are no normative references in this document.

3 Terms and definitions

ISO and IEC maintain terminology databases for use in standardization at the following addresses:

— ISO Online browsing platform: available at https://www.iso.org/obp

— IEC Electropedia: available at https://www.electropedia.org/

3.1 Terms relating to assets

3.1.1
asset
item, thing or entity that has potential or actual *value* (3.3.28) to an *organization* (3.3.1)

Note 1 to entry: Assets can be physical or non-physical.

Note 2 to entry: A grouping of assets referred to as an *asset system* (3.1.6) can also be considered as an asset.

3.1.2
asset life
period from *asset* (3.1.1) ideation to asset end of life

Note 1 to entry: "Useful life" is the period over which an asset is capable of fulfilling a purpose to an entity.

Note 2 to entry: Asset life can differ from the period the *organization* (3.3.1) holds responsibility for the asset.

Note 3 to entry: End-of-life is when the asset can no longer cause any residual obligations.

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

3.1.3
life cycle
stages of an *asset* (3.1.1) during its life

Note 1 to entry: Each stage identifies different activities and *processes* (3.3.10).

Note 2 to entry: The naming of the stages and the activities under each stage can vary in different industry sectors and is determined by the *organization* (3.3.1). An example of the naming of the stages is: conception of need, design, construct, operate and maintain, renew/replace, dispose and extinguishment of liabilities.

Note 3 to entry: The contractual liability of an entity can involve less than one or multiple life cycle stages and may be shorter than an *asset life* (3.1.2).

3.1.4
life cycle management
applying *asset management* (3.2.1) over the *life cycle* (3.1.3) of *assets* (3.1.1)

3.1.5
asset type
grouping of *assets* (3.1.1) having common characteristics that distinguish those assets as a group or class

3.1.6
asset system
set of *assets* (3.1.1) that interact or are interrelated

3.1.7
asset portfolio
assets (3.1.1) that are within the scope of an *asset management system* (3.3.5)

Note 1 to entry: A portfolio is typically established and assigned for managerial control purposes.

3.1.8
critical asset
asset (3.1.1) having the potential to significantly impact the achievement of an *organization's* (3.3.1) *objectives* (3.3.7)

Note 1 to entry: Assets can be safety-critical, environment-critical or performance-critical and can relate to legal, regulatory or statutory *requirements* (3.3.16).

Note 2 to entry: Critical assets can refer to those assets necessary to provide services to critical customers.

Note 3 to entry: *Asset systems* (3.1.6) can be distinguished as being critical in a similar manner to individual assets.

3.1.9
object, noun
anything perceivable or conceivable

Note 1 to entry: An object can be physical (e.g. a pump, a bridge, a building), or non-physical (e.g. a project plan, copyright, software).

3.1.10
data
facts about an *object* (3.1.9)

[SOURCE: ISO 9000:2015, 3.8.1]

3.1.11
information
meaningful result of organizing and processing *data* (3.1.10)

EXAMPLE Make/model/type of a pump in a specific location, span of a bridge, *asset* (3.1.1) *performance* (3.3.13) indicator, an asset's current total cost of ownership.

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

3.2 Terms relating to asset management

3.2.1
asset management
coordinated activity of an *organization* (3.3.1) to realize *value* (3.3.28) from *assets* (3.1.1)

Note 1 to entry: Realization of value normally involves a balancing of costs, *risks* (3.3.8), *opportunities* (3.3.9) and *performance* (3.3.13) benefits.

Note 2 to entry: Activity can also refer to the application of the elements of an *asset management system* (3.3.5).

Note 3 to entry: The term "activity" has a broad meaning and can include, for example, the approach, planning and plans, and their implementation.

3.2.2
organizational objective
overarching *objective* (3.3.7) that sets the context and direction for an *organization's* (3.3.1) activities

Note 1 to entry: Organizational objectives are established through the strategic level planning activities of the organization.

3.2.3
organizational plan
documented information (3.3.12) that specifies the plans to achieve the *organizational objectives* (3.2.2)

3.2.4
management
coordinated activities to direct and control an *organization* (3.3.1)

Note 1 to entry: Management can include establishing strategies, *policies* (3.3.6) and *objectives* (3.3.7), and *processes* (3.3.10) to achieve those objectives.

Note 2 to entry: Management includes the processes of planning, organizing, directing and controlling (e.g. the outcomes of people, groups or organizations).

Note 3 to entry: Control can include defining roles, appointing authority, assigning tasks, establishing incentives and rewards, and empowering and engaging people.

Note 4 to entry: The word "management" sometimes refers to people, i.e. a person or group of people with the authority and responsibility for the conduct and control of an organization. When "management" is used in this sense, it should always be used with some form of qualifier, e.g. "top management".

Note 5 to entry: The term "management" can be qualified by a specific domain it addresses. Examples include public health management, environmental management and *risk* (3.3.8) management.

[SOURCE: ISO 9000:2015, 3.3.3, modified — "strategies" added to Note 1 to entry. Note 2 to entry shortened and renumbered as Note 4 to entry. Notes 2, 3 and 5 to entry added.]

3.2.5
asset management plan
AMP
documented information (3.3.12) that specifies the activities, resources, costs and timescales required for an individual *asset* (3.1.1), or a grouping of assets, to achieve an *organization's* (3.3.1) *asset management* (3.2.1) *objectives* (3.3.7)

Note 1 to entry: The grouping of assets may be by *asset type* (3.1.5), asset class, *asset system* (3.1.6) or *asset portfolio* (3.1.7).

Note 2 to entry: An asset management plan is derived from the *strategic asset management plan* (3.3.29).

Note 3 to entry: An asset management plan may be contained in, or may be a subsidiary plan of, the strategic asset management plan.

© ISO 2024 – All rights reserved

3.2.6
capability
measure of capacity and the ability of an entity (system or *organization* (3.3.1)) to achieve its *objectives* (3.3.7)

Note 1 to entry: *Asset management* (3.2.1) capabilities include *processes* (3.3.10), resources, *competences* (3.3.11) and technologies to enable the effective and efficient development and delivery of *asset management plans* (3.2.5) and *asset life* (3.1.2) activities, and their *continual improvement* (3.3.14).

3.2.7
level of service
parameters, or combination of parameters, which reflect social, political, environmental and economic outcomes that an *organization* (3.3.1) delivers

Note 1 to entry: The parameters may include safety, customer satisfaction, quality, quantity, capacity, reliability, responsiveness, environmental acceptability, cost and availability.

3.3 Terms relating to asset management system

3.3.1
organization
person or group of people that has its own functions with responsibilities, authorities and relationships to achieve its *objectives* (3.3.7)

Note 1 to entry: The concept of organization includes, but is not limited to, sole-trader, company, corporation, firm, enterprise, authority, partnership, charity or institution, or part or combination thereof, whether incorporated or not, public or private.

Note 2 to entry: If the organization is part of a larger entity, the term "organization" refers only to the part of the larger entity that is within the scope of the *asset management system* (3.3.5).

3.3.2
stakeholder
interested party
person or *organization* (3.3.1) that can affect, be affected by, or perceive itself to be affected by a decision or activity

3.3.3
top management
person or group of people who directs and controls an *organization* (3.3.1) at the highest level

Note 1 to entry: Top management has the power to delegate authority and provide resources within the organization.

Note 2 to entry: If the scope of the *management system* (3.3.4) covers only part of an organization, then top management refers to those who direct and control that part of the organization.

3.3.4
management system
set of interrelated or interacting elements of an *organization* (3.3.1) to establish *policies* (3.3.6) and *objectives* (3.3.7), as well as *processes* (3.3.10) to achieve those objectives

Note 1 to entry: A management system can address a single discipline or several disciplines.

Note 2 to entry: The management system elements include the organization's structure, roles and responsibilities, planning and operation.

3.3.5
asset management system
management system (3.3.4) for *asset management* (3.2.1)

3.3.6
policy
intentions and direction of an *organization* (3.3.1) as formally expressed by its *top management* (3.3.3)

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

3.3.7
objective
result to be achieved

Note 1 to entry: An objective can be strategic, tactical, or operational.

Note 2 to entry: Objectives can relate to different disciplines (such as finance, health and safety, and environment). They can be, for example, organization-wide or specific to a project, service, product or *process* (3.3.10).

Note 3 to entry: An objective can be expressed in other ways, e.g. as an intended result, as a purpose, as an operational criterion, as an asset management objective or by the use of other words with similar meaning (e.g. aim, goal, or target).

Note 4 to entry: In the context of *asset management systems* (3.3.5), asset management objectives are set by the *organization* (3.3.1), consistent with the asset management *policy* (3.3.6), to achieve specific results.

3.3.8
risk
effect of uncertainty on *objectives* (3.3.7)

Note 1 to entry: An effect is a deviation from the expected — positive or negative.

Note 2 to entry: Uncertainty is the state, even partial, of deficiency of *information* (3.1.11) related to, understanding or *knowledge* (3.3.26) of, an event, its consequence, or likelihood.

Note 3 to entry: Risk is often characterized by reference to potential events (as defined in ISO 31073) and consequences (as defined in ISO 31073), or a combination of these.

Note 4 to entry: Risk is often expressed in terms of a combination of the consequences of an event (including changes in circumstances) and the associated likelihood (as defined in ISO 31073) of occurrence.

3.3.9
opportunity
circumstance or set of circumstances that can be exploited to improve *performance* (3.3.13), exceed original *objectives* (3.3.7) and/or intended results, pursue additional objectives, or achieve *continual improvement* (3.3.14)

3.3.10
process
set of interrelated or interacting activities that uses or transforms inputs to deliver a result

Note 1 to entry: Whether the result of a process is called an output, a product or a service depends on the context of the reference.

3.3.11
competence
ability to apply *knowledge* (3.3.26) and skills to achieve intended results

3.3.12
documented information
information (3.1.11) required to be controlled and maintained by an *organization* (3.3.1) and the medium on which it is contained

Note 1 to entry: Documented information can be in any format and media and from any source.

Note 2 to entry: Documented information can refer to:

— the *management system* (3.3.4), including related *processes* (3.3.10);

— information created in order for the organization to operate (documentation);

— evidence of results achieved (records).

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

3.3.13
performance
measurable result

Note 1 to entry: Performance can relate either to quantitative or qualitative findings.

Note 2 to entry: Performance can relate to managing activities, *processes* (3.3.10), products, services, systems or *organizations* (3.3.1).

Note 3 to entry: For the purposes of *asset management* (3.2.1), performance can relate to *assets* (3.1.1) in their ability to fulfil *requirements* (3.3.16) or *objectives* (3.3.7) and thereby their ability to realize *value* (3.3.28).

3.3.14
continual improvement
recurring activity to enhance *performance* (3.3.13)

3.3.15
effectiveness
extent to which planned activities are realized and planned results are achieved

3.3.16
requirement
need or expectation that is stated, generally implied or obligatory

Note 1 to entry: "Generally implied" means that it is custom or common practice for the *organization* (3.3.1) and *stakeholders* (3.3.2) that the need or expectation under consideration is implied.

Note 2 to entry: A specified requirement is one that is stated, e.g. in *documented information* (3.3.12).

3.3.17
conformity
fulfilment of a *requirement* (3.3.16)

3.3.18
nonconformity
non-fulfilment of a *requirement* (3.3.16)

Note 1 to entry: Nonconformity can be any deviation from *asset management system* (3.3.5) *requirements* (e.g. from relevant work standards, practices, procedures, legal requirements) or any failure of an *asset* (3.1.1) to meet or perform to specification.

3.3.19
corrective action
action to eliminate the cause(s) of a *nonconformity* (3.3.18) and to prevent recurrence

3.3.20
preventive action
action to eliminate the cause of a potential *nonconformity* (3.3.18) or other undesirable potential situation

Note 1 to entry: There can be more than one cause for a potential nonconformity.

Note 2 to entry: Preventive action is taken to prevent occurrence and to preserve an *asset's* (3.1.1) function whereas *corrective action* (3.3.19) is taken to prevent recurrence.

Note 3 to entry: Preventive action is normally carried out while the asset is functionally available and operable or prior to the initiation of functional failure.

Note 4 to entry: Preventive action includes the replenishment of consumables where the consumption is a functional requirement.

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

3.3.21
predictive action
action that predicts future behaviour of a parameter required to support decision-making

Note 1 to entry: Predictive actions inform *asset management* (3.2.1) decisions inclusive of preventing and/or correcting a *nonconformity* (3.3.18) as well as the realization of maximum *value* (3.3.28) from an *opportunity* (3.3.9).

Note 2 to entry: Predictive actions do not alter the state of *assets* (3.1.1) and/or systems themselves.

Note 3 to entry: Predictive actions are a precursor, but not a prerequisite, to decision-making.

Note 4 to entry: Predictive actions may include the following activities: stock market analysis, venture capital acquisition, exploration, research, investigation, experimentation, modelling, testing, surveillance, *monitoring* (3.3.25), statistical analyses, calculations, algorithm development, expert system development, artificial intelligence system development, digital twinning, alarm and warning system development.

3.3.22
incident
unplanned event or occurrence

3.3.23
audit
systematic and independent *process* (3.3.10) for obtaining evidence and evaluating it objectively to determine the extent to which the audit criteria are fulfilled

Note 1 to entry: An audit can be an internal audit (first party) or an external audit (second party or third party) and it can be a combined audit (combining two or more disciplines).

Note 2 to entry: An internal audit is conducted by the *organization* (3.3.1) itself, or by an external party on its behalf.

Note 3 to entry: "Audit evidence" and "audit criteria" are defined in ISO 19011.

3.3.24
measurement
process (3.3.10) to determine a *value* (3.3.28)

Note 1 to entry: In this definition a "value" refers to a numerical value, which is different than value in the context of *stakeholder's* (3.3.2) value or *asset management* (3.2.1) value.

3.3.25
monitoring
determining the status of a system, a *process* (3.3.10) or an activity

Note 1 to entry: To determine the status, there can be a need to check, supervise or critically observe.

Note 2 to entry: For the purposes of *asset management* (3.2.1), monitoring can also refer to determining the status of an *asset* (3.1.1). This is typically referred to as "condition monitoring" or "performance monitoring".

3.3.26
knowledge
understanding of facts, information and principles acquired through experience, research or education

Note 1 to entry: Asset management knowledge can be derived from:

a) internal sources (e.g. intellectual property; knowledge gained from experience; lessons learned from failures and successful initiatives; bodies of knowledge; informal sources of knowledge and experience of people; the results of improvements in *processes* (3.3.10), products and services; simulation models); reference can be made to ISO 30401 for additional guidance;

b) external sources (e.g. standards; books, academia; conferences; gathering knowledge from customers, collaborators or external providers).

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

3.3.27
sustainability
state of the global system, including environmental, social and economic aspects, in which the needs of the present are met without compromising the ability of future generations to meet their own needs

Note 1 to entry: The environmental, social and economic aspects interact, are interdependent and are often referred to as the three dimensions of sustainability.

Note 2 to entry: Sustainability is the *objective* (3.3.7) of sustainable development.

Note 3 to entry: In *asset management* (3.2.1), sustainability of *assets* (3.1.1) or *asset systems* (3.1.6) refer to their ability to meet present *objectives* without compromising their ability to meet future *objectives*.

[SOURCE: ISO Guide 82:2019, 3.1, modified — In Note 2 to entry, "goal" has been replaced by "objective". Note 3 to entry has been added.]

3.3.28
value
results from satisfying needs and expectations

Note 1 to entry: Value represents the result of considering positive and negative impacts, as well as financial and non-financial impacts, on *stakeholders* (3.3.2) over a time horizon that includes all such impacts.

3.3.29
strategic asset management plan
SAMP
documented information (3.3.12) that contains and aligns *asset management* (3.2.1) *policy* (3.3.6), *objectives* (3.3.7), strategies and approaches for developing and managing the *asset portfolio* (3.1.7) and the *asset management system* (3.3.5)

Note 1 to entry: A SAMP is derived from, may be contained in, or may be a subsidiary plan of, the *organizational plan* (3.2.3).

3.3.30
framework
structure of *processes* (3.3.10) and specifications designed to support the accomplishment of a specific task

Note 1 to entry: The framework for *asset management* (3.2.1) decision-making is used by the *organization* (3.3.1):

— in defining and determining the *value* (3.3.28) that it aims to derive from its *assets* (3.1.1) by applying asset management;

— in defining the criteria to be used for asset management decision-making to achieve value.

4 Overview of asset management

4.1 General

Asset management is a crucial business activity that involves various disciplines. It applies to any organization, at all levels and across all departments, as each contributes to delivering value from the portfolio of assets. It's important for everyone in the organization and its value chain to understand the organization's purpose, objectives and how asset management contributes.

Asset management often involves resolving conflicting objectives, risks, opportunities, costs and other impacts on different stakeholders over varying time frames. Decision-making in asset management must therefore address trade-offs to determine the best value options and priorities. Asset management enables organizations to assess the need for, and the performance of, assets at different stages of their life cycles.

Since long-lived assets span multiple changes in organizational leadership, the realization of value over the life cycle requires a long-term perspective and consistent stewardship aligned with organizational objectives. Effective asset management contributes to organizational objectives and sustainability by planning for, and creating and preserving value from, the total asset life cycle.

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

To establish effective asset management, an organization should apply and adopt certain fundamental principles. These, consolidated within a structured asset management system, ensure that there is consistent focus on the right things to do and for the right reasons, and that these are delivered in a coordinated way. Such an approach will result in improved outcomes and benefits.

4.2 Principles of asset management

4.2.1 General

Principles are fundamental guidelines for decision-making, behaviour and actions within an organization. They serve as the foundation for the organization's values and culture. Principles are applied across various aspects of an organization, such as governance, ethics and operations. They help in establishing a consistent approach to addressing challenges and making choices. The key principles of asset management are given in 4.2.2 to 4.2.4.

4.2.2 Value

Asset management focuses on the value assets provide to the organization over time. This principle encourages clear contributions to organizational objectives, life cycle management and transparent decision-making processes.

4.2.3 Alignment

Asset management aligns financial, technical and operational decisions with the organizational objectives, promoting vertical and horizontal coordination. Vertical alignment links organizational purpose, asset management objectives, strategies, plans and activities, while horizontal alignment represents collaboration across departments, functions and responsibilities for different stages of the asset life cycle.

4.2.4 Leadership

Leadership and sustained commitment at all levels are crucial for successful asset management. Effective leadership establishes a clear direction, roles, responsibilities and authorities, with transparent consultation and the development of a collaborative culture that is aligned to organizational success.

4.3 Outcomes and benefits from asset management

The primary outcomes of asset management are the realization of value and the achievement of organizational objectives. There are also some characteristics that result from the implementation of a systematic approach to asset management, including the following:

— Assurance: Asset management provides better organizational oversight and accountability. It assures that the right decisions are being made, the right resources and activities applied, and that assets can fulfil their required purposes over the time frames of such requirements.

— Adaptability: Asset management enables organizations to adapt more rapidly and effectively to changes in the internal and external context. This is achieved through monitoring internal and external environments, stakeholder involvement and responsive business cycles in planning and reporting.

— Sustainability: Asset management promotes long-term thinking, considering future impacts, risks, uncertainties and opportunities to realize value. It considers environmental, social and economic sustainability, spanning the entire asset life cycle. Sustainability is achieved by looking beyond short-term needs and priorities to consider future impacts and consequences, in order to develop and deliver sustainable plans and activities.

The more specific benefits of asset management include, but are not limited to, the following:

a) Improved financial performance: Boosting return on investments and reducing costs without compromising organizational objectives in the short or long term.

© ISO 2024 – All rights reserved

b) Better decision-making: Greater rigour, clarity and consistency in how decisions are made, in the use of data to support these decisions and in the value represented by options that are chosen.

c) Managed risk and opportunity: Systematically identifying, evaluating, and controlling risks and opportunities, ensuring appropriate financial outcomes, and enhancing asset and service resilience, occupational health and safety, and environmental and social impact.

d) Greater stakeholder satisfaction and confidence: Ensuring the performance of assets, leading to better service delivery that consistently meets customer and other stakeholder expectations.

e) More evident corporate social responsibility: Enhancing the organization's ability to adopt policies, ethical business practices, and stewardship and activities that promote social and environmental well-being.

f) Demonstrated compliance: An asset management system improves the ability to show adherence to legal, statutory and regulatory requirements, as well as complying with asset management standards, policies, processes and procedures.

g) Enhanced reputation: Effective asset management can improve customer satisfaction, public confidence and other stakeholder perceptions through demonstrating responsible stewardship of assets and resources.

h) Improved efficiency and effectiveness: Improving asset management planning, processes, procedures and asset performance produces greater efficiency and effectiveness of activities. This includes better resource utilization, risk control, teamwork, employee engagement and productivity.

i) Improved co-ordination and communication: Asset management results in better collaboration, understanding, behaviours and alignment across all areas of the organization, including support functions and value chain.

j) Increased value from innovation: The clearer and more consistent focus on value helps in the appropriate selection, use and benefits obtained from new methods, ideas and technologies.

4.4 Relationship between asset management, asset management system and assets

Many organizations understand that their assets are crucial for their business success. Assets provide value when they contribute to the organization's objectives and are managed according to its strategies and policies. Organizations use an asset management system to direct, coordinate and control their asset management activities to facilitate this. Asset management improves risk control and ensures that organizational objectives are consistently achieved. Adopting asset management principles is essential for an effective asset management culture.

The relationship between asset management, the organization's asset management system and the assets it manages is shown in Figure 1.

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

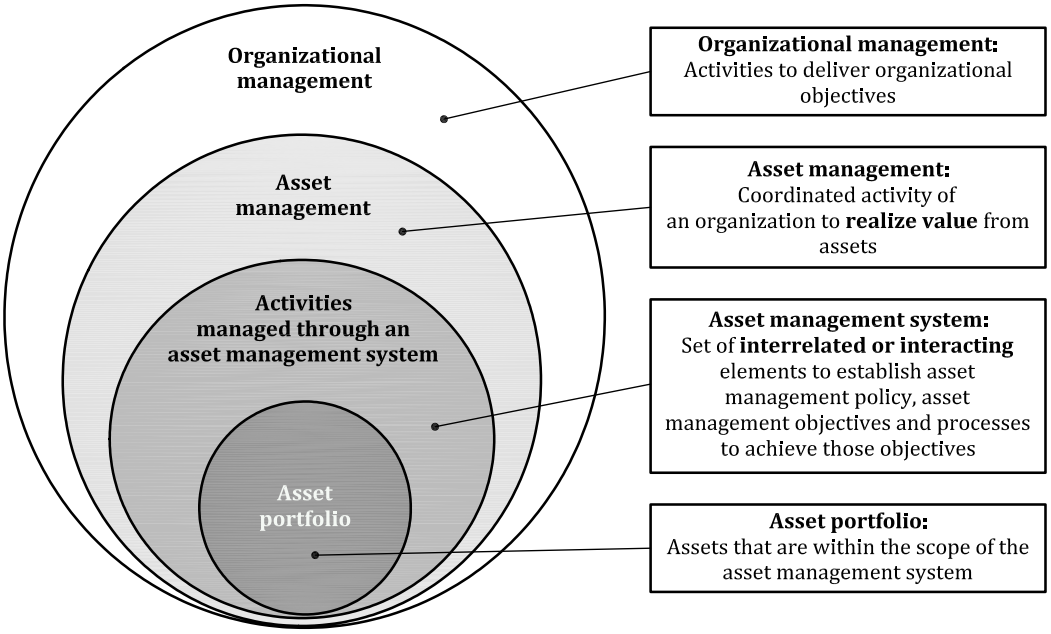

Figure 1 — Relationship between key terms in asset management

ISO 55001 and ISO 55002 specify requirements and give guidance on how to apply the asset management system.

Management systems can be organized in different ways, either informally or with a more formal structure, depending on the organization's specific needs and operational complexity. Many organizations find incorporating these asset management system requirements into their overall management system beneficial. This integration helps them improve the efficiency and effectiveness of their management activities.

The Harmonized Structure, which serves as the basis for all ISO Management System Standards, facilitates this integrated approach. An asset management system can become part of the organization's broader management framework by aligning the requirements of various management systems. Moreover, it improves process management and makes it easier for the organization to accept each new system.

As an example, if an organization already uses a risk management approach in its management system, it can be adapted to include asset management risks. This integrated approach speeds up the benefits derived from asset management and an asset management system.

4.5 Asset management improvement and maturity

Asset management maturity is key to enhancing an organization's performance. The organization's progress can be measured by its maturity: showing how well it is performing and where targeted improvements can be made. This involves consistently developing asset management capabilities, improving the asset management system and its implementation, and adopting the principles. This can be considered as part of the continual improvement process.

An organization's asset management and asset management system maturity are reflected in its capabilities, which are essential elements enabling success. The specific type and strength of these capabilities depend on the organization's context and objectives. Organizations must decide whether there is identifiable value in further improvement or aiming for excellence. Mature organizations consider the changing nature of their operating environments, recognizing opportunities from internal and external sources that can enhance performance if managed effectively.

Indicators of an organization's asset management maturity include:

— understanding and communicating the value the organization wants from its assets and incorporating it in the planning processes and management reviews;

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

— adopting a complete life cycle perspective for all types of assets;

— accurate information about its assets for future use and decision-making;

— integration of the asset management system into the overall organization's management system;

— effective management of risks and opportunities, particularly understanding the services, assets, capabilities and processes to achieve its objectives;

— awareness of the external context and potential changes, setting appropriate objectives, understanding necessary capabilities and having timely plans to achieve these;

— a supportive asset management culture;

— integration and collaboration across the organization, especially in setting priorities and making compromises when needed;

— established continual improvement.

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Bibliography

[1] ISO 9000:2015, *Quality management systems — Fundamentals and vocabulary*

[2] ISO 19011, *Guidelines for auditing management systems*

[3] ISO 30401, *Knowledge management systems — Requirements*

[4] ISO 31073, *Risk management — Vocabulary*

[5] ISO 55001, *Asset management — Asset management system — Requirements*

[6] ISO 55002, *Asset management — Management systems — Guidelines for the application of ISO 55001*

[7] ISO/TS 55010, *Asset management — Guidance on the alignment of financial and non-financial functions in asset management*

[8] ISO 55011, *Asset management — Guidance for the development of public policy to enable asset management*

[9] ISO 55012, *Asset management — Guidance on people involvement and competence*

[10] ISO 55013, *Asset management — Guidance on the management of data assets*

[11] ISO Guide 82:2019, *Guidelines for addressing sustainability in standards*

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Index

A

AMP 3.2.5
asset 3.1.1
asset life 3.1.2
asset management 3.2.1
asset management plan 3.2.5
asset management system 3.3.5
asset portfolio 3.1.7
asset system 3.1.6
asset type 3.1.5
audit 3.3.23

C

capability 3.2.6
competence 3.3.11
conformity 3.3.17
continual improvement 3.3.14
corrective action 3.3.19
critical asset 3.1.8

D

data 3.1.10
documented information 3.3.12

E

effectiveness 3.3.15

F

framework 3.3.30

I

incident 3.3.22
information 3.1.11
interested party 3.3.2

K

knowledge 3.3.26

L

level of service 3.2.7
life cycle 3.1.3
life cycle management 3.1.4

M

management 3.2.4
management system 3.3.4
measurement 3.3.24
monitoring 3.3.25

N

nonconformity 3.3.18

O

object, noun 3.1.9
objective 3.3.7
opportunity 3.3.9
organization 3.3.1
organizational objective 3.2.2
organizational plan 3.2.3

P

performance 3.3.13
policy 3.3.6
predictive action 3.3.21
preventive action 3.3.20
process 3.3.10

R

requirement 3.3.16
risk 3.3.8

S

SAMP 3.3.29
stakeholder 3.3.2
strategic asset management plan 3.3.29
sustainability 3.3.27

T

top management 3.3.3

V

value 3.3.28

© ISO 2024 – All rights reserved

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

This page deliberately left blank

NO COPYING WITHOUT BSI PERMISSION EXCEPT AS PERMITTED BY COPYRIGHT LAW

Licenced to Matthew Jackson, Devonport Royal Dockyard Limited downloaded on 19/07/2024. For status of this document please check https://knowledge.bsigroup.com. Reproduction and distribution is prohibited.

British Standards Institution (BSI)

BSI is the national body responsible for preparing British Standards and other standards-related publications, information and services.

BSI is incorporated by Royal Charter. British Standards and other standardization products are published by BSI Standards Limited.

About us

We bring together business, industry, government, consumers, innovators and others to shape their combined experience and expertise into standards -based solutions.

The knowledge embodied in our standards has been carefully assembled in a dependable format and refined through our open consultation process. Organizations of all sizes and across all sectors choose standards to help them achieve their goals.

Information on standards

We can provide you with the knowledge that your organization needs to succeed. Find out more about British Standards by visiting our website at bsigroup.com/standards or contacting our Customer Services team or Knowledge Centre.

Buying standards

You can buy and download PDF versions of BSI publications, including British and adopted European and international standards, through our website at bsigroup.com/shop, where hard copies can also be purchased.

If you need international and foreign standards from other Standards Development Organizations, hard copies can be ordered from our Customer Services team.

Copyright in BSI publications

All the content in BSI publications, including British Standards, is the property of and copyrighted by BSI or some person or entity that owns copyright in the information used (such as the international standardization bodies) and has formally licensed such information to BSI for commercial publication and use.

Save for the provisions below, you may not transfer, share or disseminate any portion of the standard to any other person. You may not adapt, distribute, commercially exploit or publicly display the standard or any portion thereof in any manner whatsoever without BSI's prior written consent.

Storing and using standards

Standards purchased in soft copy format:

- A British Standard purchased in soft copy format is licensed to a sole named user for personal or internal company use only.
- The standard may be stored on more than one device provided that it is accessible by the sole named user only and that only one copy is accessed at any one time.
- A single paper copy may be printed for personal or internal company use only.

Standards purchased in hard copy format:

- A British Standard purchased in hard copy format is for personal or internal company use only.
- It may not be further reproduced – in any format – to create an additional copy. This includes scanning of the document.

If you need more than one copy of the document, or if you wish to share the document on an internal network, you can save money by choosing a subscription product (see 'Subscriptions').

Reproducing extracts

For permission to reproduce content from BSI publications contact the BSI Copyright and Licensing team.

Subscriptions

Our range of subscription services are designed to make using standards easier for you. For further information on our subscription products go to bsigroup.com/subscriptions.

With **British Standards Online (BSOL)** you'll have instant access to over 55,000 British and adopted European and international standards from your desktop. It's available 24/7 and is refreshed daily so you'll always be up to date.

You can keep in touch with standards developments and receive substantial discounts on the purchase price of standards, both in single copy and subscription format, by becoming a **BSI Subscribing Member.**

PLUS is an updating service exclusive to BSI Subscribing Members. You will automatically receive the latest hard copy of your standards when they're revised or replaced.

To find out more about becoming a BSI Subscribing Member and the benefits of membership, please visit bsigroup.com/shop.

With a **Multi-User Network Licence (MUNL)** you are able to host standards publications on your intranet. Licences can cover as few or as many users as you wish. With updates supplied as soon as they're available, you can be sure your documentation is current. For further information, email cservices@bsigroup.com.

Revisions

Our British Standards and other publications are updated by amendment or revision.

We continually improve the quality of our products and services to benefit your business. If you find an inaccuracy or ambiguity within a British Standard or other BSI publication please inform the Knowledge Centre.

Useful Contacts

Customer Services
Tel: +44 345 086 9001
Email: cservices@bsigroup.com

Subscriptions
Tel: +44 345 086 9001
Email: subscriptions@bsigroup.com

Knowledge Centre
Tel: +44 20 8996 7004
Email: knowledgecentre@bsigroup.com

Copyright & Licensing
Tel: +44 20 8996 7070
Email: copyright@bsigroup.com

BSI Group Headquarters

389 Chiswick High Road London W4 4AL UK